SAGT: Structure-Adaptive Graph Transformer for Hyperspectral Image Classification

Shuyu Zhang, *Member, IEEE*, Shuguo Jiang, Wenlong Yin, Weixi Wang, Meng Xu, *Member, IEEE*, Jiasong Zhu, and Sen Jia, *Senior Member, IEEE*

***Abstract*—Hyperspectral images (HSIs) are vital for scene analysis, as they capture detailed spatial and spectral information to characterize surface materials. However, accurate HSI classification is challenged by significant intraclass spectral variability and spatial complexity. To address this, we leverage the fact that pixels of the same class typically form irregular local regions. We propose a structure-adaptive graph Transformer (SAGT) that dynamically captures irregular spatial topologies and homogeneous spectral information to achieve adaptive HSI representation and precise classification. Specifically, a structure-aware self-attention (SASA) module is developed to embed graph structures into the self-attention (SA) mechanism as a robust positional indicator, which can be extended easily and effectively. SASA comprehensively accounts for the spatial structures and spectral autocorrelation of ground objects, facilitating the aggregation of homogeneous spectral information for noise-robust spectral representations. Additionally, a structure-adaptive pooling (SAP) module is designed to dynamically adjust graph structures by discarding irrelevant edges, thus better indicating spatial relationships. By coupling the SASA and SAP modules, our proposed SAGT model significantly alleviates spectral variability and tolerates prior noise. Furthermore, data augmentation techniques of random discard and random offset are built, which randomly drop and shift graph nodes to generate more diverse samples during preprocessing. In post-processing, multiview decision-making integrates results from multiple contextual views to provide more robust predictions. Experimental results on three benchmark datasets consistently demonstrate that SAGT is more effective and reliable than other state-of-the-art methods. To facilitate reproduction, we will release the source code for SAGT at https://github.com/ShuGuoJ/SAGT.git**

***Index Terms*—Data augmentation, hyperspectral image (HSI) classification, multiview decision, structure-adaptive graph Transformer (SAGT).**

I. Introduction

HYPERSPECTRAL imaging technology has made significant progress in recent years, leading to substantial improvements in the spectral resolution of observations. Simultaneously, the availability of various platforms, such as uncrewed aerial vehicles (UAVs) and satellites, has made hyperspectral images (HSIs) increasingly accessible. HSIs are widely used in earth remote sensing, environmental monitoring, and object tracking, providing rich spatial geometries and distinct spectral responses. Similar to semantic segmentation in natural images, HSI classification aims to accurately identify the categories of all pixels based on their unique spectral–spatial characteristics, particularly for objects that are challenging to distinguish by color, appearance, or texture. However, HSI classification often encounters issues like spectral variability, high dimensionality, and mixed pixels, which can lead to misclassification and suboptimal performance [1], [2]. Traditional machine learning methods can extract features and recognize types through manual design, but they lack the capability for automatic feature learning and high-level semantic abstraction.

In the past decade, deep learning has achieved remarkable success in HSI classification [3], [4]. Unlike traditional machine learning, deep learning can automatically learn complex and effective patterns from high-dimensional data. Most of the current deep learning methods for HSI classification [5], [6], [7], [8], [9], [10], [11], [12], [13] are based on convolutional neural networks (CNNs), which apply multiple convolution filters across the image to capture semantic features layer by layer [14], [15], [16], [17], [18], [19]. Although CNNs have the advantages of locality, translation invariance, and shared weights, their receptive fields are limited by the size of the convolution filters, making it difficult to capture long-range feature dependencies. Additionally, CNNs operate on Euclidean data with patch inputs, limiting their ability to generalize to more flexible HSI formats, such as graphs.

Recently, the Transformer, originally developed for NLP, has emerged as an alternative to CNNs in computer vision. Its core component of self-attention (SA) allows models to focus on internal dependencies and efficiently aggregate global

Received 8 July 2025; revised 4 November 2025; accepted 12 November 2025. Date of publication 18 November 2025; date of current version 4 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62201354 and Grant 62271327; in part by Shenzhen Science and Technology Program under Grant JCYJ20240813142308012, Grant RCJC20221008092731042, Grant JCYJ20220818100206015, Grant KQTD20200909113951005, and Grant KCXFZ20240903093100001; in part by the Project of Department of Education of Guangdong Province under Grant 2023KCXTD029; in part by the Research Team Cultivation Program of Shenzhen University under Grant 2023JCT002; in part by the Scientific Foundation for Youth Scholars of Shenzhen University under Grant 827-0001080; and in part by the Open Fund of State Key Laboratory of Remote Sensing and Digital Earth under Grant OFSLRSS202510. *(Corresponding author: Sen Jia.)*

Shuyu Zhang, Wenlong Yin, Meng Xu, and Sen Jia are with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518060, China (e-mail: shuyu-zhang@szu.edu.cn; 2310273019@email.szu.edu.cn; m.xu@szu.edu.cn; senjia@szu.edu.cn).

Shuguo Jiang is with the School of Computer Science, Wuhan University, Wuhan 430072, China (e-mail: shuguoj@foxmail.com).

Weixi Wang is with the College of Architecture and Urban Planning, Shenzhen University, Shenzhen 518060, China (e-mail: wangwx@szu.edu.cn).

Jiasong Zhu is with the Key Laboratory for Geo-Environmental Monitoring of Coastal Zone of the Ministry of Natural Resources, Shenzhen University, Shenzhen 518060, China (e-mail: zhujiasong@gmail.com).

Digital Object Identifier 10.1109/TGRS.2025.3634114

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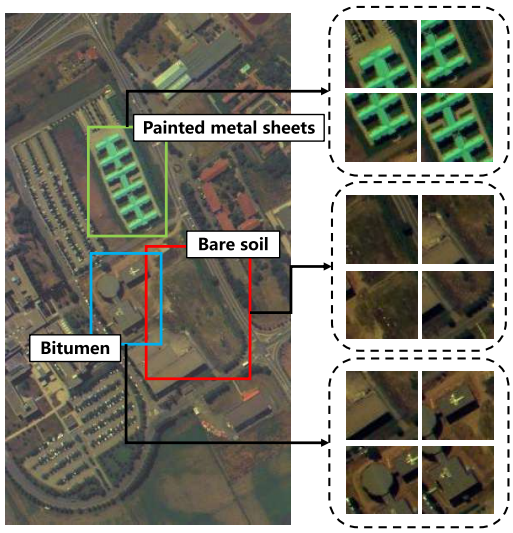


Fig. 1. In hyperspectral remote sensing images, the same ground object can exhibit different spatial geometries in local patches, such as painted metal sheets, while different ground objects, like bare soil and bitumen, may share similar local spatial patterns.

information. Vision Transformer (ViT) [20], the first pure Transformer for image classification [21], [22], [23], achieved results comparable to CNN-based methods. There has been a surge of Transformer-based methods [24], [25], [26], [27], [28], [29], [30], [31] for HSI classification, primarily due to their ability to model long-range relationships. These methods have explored long-range interactions between pixels [32], patches [33], and spectral bands [34], achieving promising results. At the same time, significant efforts have been made to process data formats like graphs, which are more flexible and complex than Euclidean data. Popular graph convolutional networks (GCNs) [35], [36], [37], [38], [39] perform convolution on graphs for feature extraction. The emergence of GCNs has extended deep learning from images to the non-Euclidean graph domain, enabling the representation of complex relationships and the extraction of topological structures. This type of HSI classification methods [40], [41], [42] emphasize appropriate graph construction to capture spatial relationships at pixel level [43], [44], superpixel level [45], [46], and in mixed scenarios [47]. By leveraging the flexible representation of graphs, these methods can mitigate background interference and enhance classification accuracy.

The spectral variability in HSIs, where different objects may have similar spectra and the same objects may have different spectra, limits the effectiveness of spectral-based methods. Although spatial geometries can help mitigate this issue, there is often a weak and variable correlation between ground objects and their local spatial structures in patches, as illustrated in Fig. 1. From our perspective, incorporating spatial information involves smoothing the spectral differences of the same ground objects through local homogeneous pixels, thereby reducing intraclass variability and enhancing interclass differences. Transformers are particularly effective at adaptively extracting homogeneous HSI information through a "query-key-token" mechanism, but they are less sensitive to positional changes, necessitating additional positional encoding. Vaswani et al. [21] introduced a sinusoidal function to indicate element positions in sequences, which Dosovitskiy et al. [20] later replaced with an array of learnable model parameters. To optimize absolute positional encoding, Liu et al. [48] proposed a positional bias that stores relative position information in a predefined matrix. In practice, input sizes should be adjusted according to sensor hardware, spatial resolution, and object distribution. However, current positional encoding methods suffer from positional distortion when applied beyond their training range, leading to inaccurate spatial relationships and degraded HSI classification performance. Consequently, designing a robust, plug-and-play, and effectively generalizable positional encoding for HSI inference remains a challenge.

On the other hand, graph structures are powerful for representing complex and relative relationships, with the ability to expand freely and preserve the original topology. This makes graphs a promising candidate for serving as robust positional indicators for Transformers without the need to predefine the span. However, the graph structure in HSIs is not explicit, and hand-crafted graphs used in [43] and [45] lack flexibility and often contain prior noise. Hence, learning node representations and constructing graph structures independently can lead to suboptimal results in graph-based HSI classification. The contextual relationships between different parts of an HSI are complex and may vary depending on the scene. If the graph structure is predefined and static, it may not adapt to changes in the spatial or spectral context, leading to rigid and possibly incorrect representations. This is particularly problematic in HSIs, where spectral variability and mixed pixels require a more dynamic approach. Furthermore, independently constructed graphs might include irrelevant or noisy edges that do not correspond to meaningful relationships in HSIs. Without a joint optimization process, it is difficult to filter out or adjust for this noise. Overall, it is highly desirable to design a network capable of capturing homogeneous features, perceiving positional changes, and dynamically adjusting graph structures in an end-to-end manner.

To address the aforementioned challenges, we propose a novel structure-adaptive graph Transformer (SAGT) for adaptive HSI representation and accurate classification, as shown in Fig. 2. First, a structure-aware SA (SASA) module is designed to embed the graph structure into the SA mechanism, where the graph structure serves as a form of relative positional encoding, enhancing spatial indications. Unlike traditional positional encoding methods, our graph-based encoding freely expresses the relative positions within the HSI and is highly scalable, allowing new positional nodes to be seamlessly integrated into the original graph. Next, a structure-adaptive pooling (SAP) module is designed to dynamically adjust the graph structure and mitigate the impact of prior noise. This is achieved by discarding edges with low similarity, allowing for iterative optimization of the graph's representation. Finally, the SASA and SAP modules are integrated within our SAGT model, enabling end-to-end training for graph construction, node learning, feature extraction, and category recognition in HSIs. In addition, we design two kinds of data augmentation techniques for graphs, called random discard and

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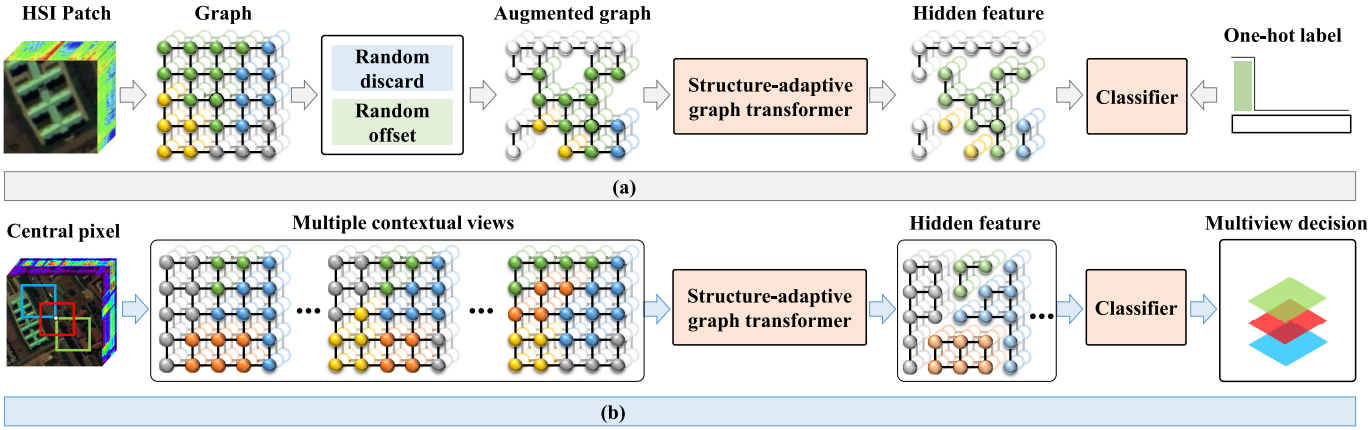


Fig. 2. Overall flowchart of our proposed SAGT method. During the training flow, an HSI patch is converted into a graph and augmented via random discard and random offset strategies to improve feature learning and structural robustness. During the inferring flow, the central pixel is analyzed through multiple contextual views and a multiview decision to integrate the results to produce the final robust classification output. (a) Training flow with preprocessing of graph augmentation. (b) Inferring flow with postprocessing of multiview integration.

random offset, which enhance sample diversity. During inference, a multiview decision-making process is developed to incorporate HSI classification results from different contextual perspectives, resulting in more robust predictions. Our method introduces three main innovations, as follows.

1) We propose a new operator, called SASA, to better represent internal image structures and aggregate homogeneous pixel features in HSIs. In SASA, the graph serves as a strong positional indicator and is embedded into the SA mechanism to enhance its sensitivity to positional changes. Notably, the graph-based positional indicator introduces no additional training parameters and is scalable, adapting to inputs of various sizes during inference. From another perspective, it integrates dynamic global content with static local positions, enabling the model to focus on essential multiscale features.
2) Considering the prior noise in hand-crafted graphs, an SAP mechanism is designed to integrate graph structure search and node representation learning. SAP can adaptively adjust the graph structure by discarding edges with low similarity, while simultaneously redirecting positional information to better aggregate homogeneous features, particularly for boundary pixels. Furthermore, we develop the SAGT model by coupling the SASA and SAP modules, offering scalability, noise resistance, and high perceptive capabilities. The deep HSI feature extraction of the Transformer and the automatic structure adjustment via graph indicator are learned end-to-end, making our SAGT both robust and flexible in a variety of situations.
3) Two types of data augmentation strategies, random discard and offset, are proposed to enhance the diversity of graph structures by randomly dropping and shifting nodes with a certain probability before training. This encourages the SAP to better distinguish between similar and dissimilar nodes and to identify appropriate homogeneous regions. During inference, a multiview decision-making process is introduced to evaluate pixels from different contextual perspectives. This approach helps gather information from multiple views about ground objects, leading to robust recognition with more complete attributes. The effectiveness and superiority of our proposed modules and model have been demonstrated through a series of experiments on benchmark datasets.

The remainder of this article is organized as follows. Section I reviews the related works on Transformer- and graph-based HSI classification. Next, the details of our proposed modules are presented in Section III, followed by the experimental results in Section IV. Finally, the conclusions are drawn in Section V.

II. Related Works

A. Transformer-Based HSI Classification

The Transformer model [21] and its variants have dominated NLP for several years due to their ability to compute efficiently in parallel and effectively capture long-range dependencies. The core component of Transformers is SA, which takes the query, key, and value matrices as inputs and outputs the attention result, as shown in

$$\mathrm{SA}(\mathbf{X}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_{\mathbf{K}}}}\right)\mathbf{V} \tag{1}$$

and

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_{\mathbf{Q}}, \mathbf{K} = \mathbf{X}\mathbf{W}_{\mathbf{K}}, \mathbf{V} = \mathbf{X}\mathbf{W}_{\mathbf{V}} \tag{2}$$

where $\mathbf{X}$ is the feature matrix and $d_{\mathbf{K}}$ is the channel dimension. $\mathbf{W}_{\mathbf{Q}}$, $\mathbf{W}_{\mathbf{K}}$, and $\mathbf{W}_{\mathbf{V}}$ are corresponding query, key, and value parameters, respectively. Nevertheless, while Transformers are effective at capturing content, they are not sensitive to positional changes. To address this, Vaswani et al. [21] introduced positional encoding using a sine–cosine function added to $\mathbf{X}$ before SA. Later, Dosovitskiy et al. [20] replaced this with an array of learnable parameters. Both approaches encoded absolute positions for all elements, which reduced their sensitivity to relative positional information. In response, Liu et al. [48] proposed incorporating a relative bias that stores positional

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information in a predefined matrix. For computer vision, the ViT [20] was the first pure Transformer model proposed for image classification and achieved results comparable to CNN-based methods. Images are divided into sequences of 16×16 tokens, which are then input into the Transformer for feature learning. Many subsequent works [48], [49], [50] have followed this approach, introducing different inductive biases into Transformers. The Swin Transformer [48], for example, modifies global attention into window-based attention and enables communication across neighboring windows by shifting.

These developments have also inspired researchers to advance HSI classification using Transformers [26], [27], [32], [33], [34], addressing the problems from both spectral and spatial perspectives. He et al. [33] and Hong et al. [34] treated spectral maps as tokens, inputting them into Transformers to capture dependencies between spectra. On the other hand, Sun et al. [26] and Xue et al. [27] employed Transformers to focus on spatial dependencies between tokens. Each approach has its own advantages and limitations for HSI classification. As previously mentioned, Transformers are not inherently sensitive to positional changes and typically use sinusoidal functions or arrays of learnable parameters for positional encoding. This limitation affects their flexibility and compatibility with HSI input patches of varying sizes, which is crucial during inference due to the different spatial resolutions and object distributions encountered in practice. Additionally, there is a need to enhance the representation and analysis of multiscale structures in HSI data, beyond the global content captured by Transformers. Consequently, designing a new and effective positional indicator and developing a precise spectral–spatial learning model are essential.

B. Graph-Based HSI Classification

Graphs are flexible and powerful tools for representing the relationships among nodes. Recently, GCNs have gained popularity for processing graphs, employing two main techniques: spectral graph convolution [35], [36] and spatial graph convolution [37], [38], [39]. Spectral graph convolution involves transforming the graph signal $\boldsymbol{x}$ from the spatial domain to the spectral domain using the Fourier transformation and then transforming it back to the spatial domain after filtering. This process can be formulated as

$$\boldsymbol{x} *_G \boldsymbol{g} = \mathbf{U}\boldsymbol{g}_\theta \mathbf{U}^\top \boldsymbol{x} \tag{3}$$

where $\mathbf{U}$ is orthogonal Fourier basis. In addition, $\boldsymbol{g}_\theta = \mathbf{U}^\top \boldsymbol{g}$, and $\boldsymbol{g}$ is another graph signal. Considering multidimensional graph signals in practice, Estrach et al. [35] replaced $\mathbf{g}_\theta$ with a series of learnable matrices $\mathbf{\Theta}_{ij}^l$ and extended it to a multidimensional formula, that

$$\boldsymbol{h}_j^{l+1} = \sum_{i=1}^{f_l} \mathbf{U}\mathbf{\Theta}_{ij}^l \mathbf{U}^\top \boldsymbol{h}_i^l,\ j = 1, \ldots, f_{l+1} \tag{4}$$

where $\boldsymbol{h}_j^{l+1}$ denotes the graph convolution at the lth layer and the jth channel, and f_{l+1} is the number of channels. This multidimensional computation forms the basis for graph convolution. Although Defferrard et al. [36] sought to simplify it using the Chebyshev expansion, the approach remained inefficient and inflexible until Kipf and Welling [37] unified spectral and spatial graph convolution through

$$\mathbf{H}^{l+1} = \widetilde{\mathbf{D}}^{-1/2}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-1/2}\mathbf{H}^l\mathbf{\Theta}^l \tag{5}$$

where $\widetilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ and $\widetilde{\mathbf{D}} = \mathbf{D} + \mathbf{I}$. Here, $\mathbf{A}$ is the adjacent matrix, $\mathbf{D}$ is the degree matrix, and $\mathbf{I}$ is the identity matrix. In comparison to spectral graph convolution, spatial graph convolution is easy to understand and extend, so many excellent works [37], [38], [39] have been produced.

The success of GCNs has also drawn attention in the field of HSI classification, leading to several related works [43], [45], [51], [52], [53], [54]. Unlike social networks and knowledge graphs, the graph structure in HSIs is not explicit, as ground objects are represented and hidden within regular pixels. Therefore, designing an effective graph structure for HSIs remains a critical challenge. Qin et al. [43] first addressed this by constructing a graph for each pixel using K-nearest neighbors, a method that is both inefficient and memory-intensive. Later works [51], [52], [53] introduced graph construction in patches using the same approach, reducing the resource demand for processing. While these methods built pixel-level graphs for HSIs, Wan et al. [45] and [46] proposed superpixel-level graphs based on HSI superpixel segmentation. Both pixel-level and superpixel-level graph construction are common techniques for HSIs, but most approaches treat graph construction and node representation as separate tasks. Additionally, hand-crafted graphs often incorporate prior noise based on subjective considerations and varying conditions, highlighting the need for graph structures to be optimized automatically and adaptively.

III. Methodology

Increasing spatial information involves smoothing intraclass spectral differences using surrounding homogeneous pixels while simultaneously enhancing interclass distinctions. To address this, we propose SASA to make SA more flexible by using a graph as a robust positional indicator and local descriptor. Additionally, SAP is designed to adapt the graph structure during propagation and reduce prior noise in hand-crafted graphs. SASA and SAP are then coupled to form the SAGT, as shown in Fig. 3, enabling end-to-end learning of graph structures, node representations, and feature relationships. To complement graph-level representations with fine-grained spatial details, SAGT incorporates a CNN branch. This branch consists of two cascaded convolutional blocks. The first block uses pointwise and depthwise convolutions for channel compression and efficient local feature extraction. The second block employs parallel 1×1 and 3×3 convolutions to capture multiscale information, enhanced by a residual connection to stabilize training and preserve feature integrity. The resulting features are transformed into an image-level representation via global pooling and fully connected layers. Then, the CNN and graph branch features are adaptively fused using a learnable balance factor, and the combined representation is passed to the classifier for prediction. Furthermore, random discard and random offset in preprocessing, along with

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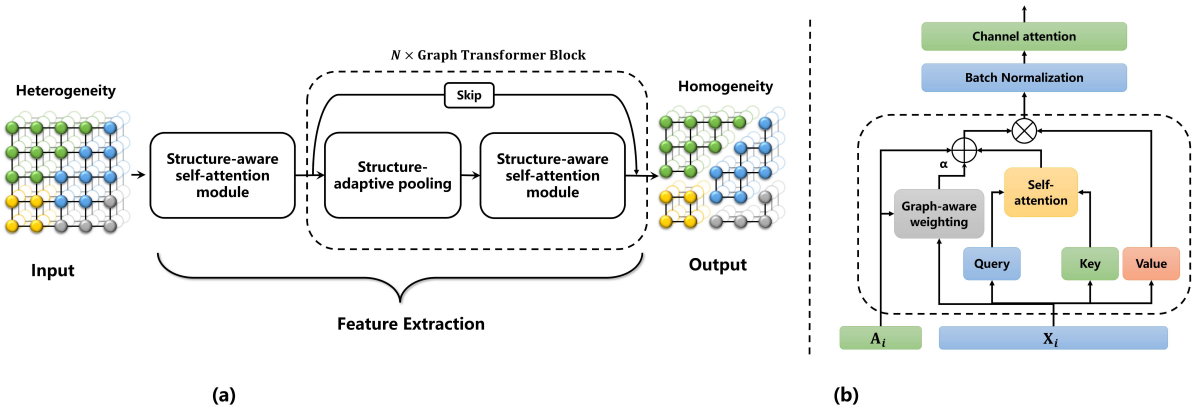


Fig. 3. (a) (Left) SAGT, which consistsof a SASA module and multiple graph Transformer blocks. It processes an input graph to output homogeneous subgraphs and their corresponding node features. (b) (Right) Specific operation of the SASA module, which embeds HSI graph structures into SA for spatial relationship perception and augments features via channel attention.

multiview decision-making in postprocessing, are designed to further improve classification performance.

In data preprocessing, an HSI with height H and width W is padded with its mirror and then is cut into $H \times W$ patches of size $M = h \times w$. The set of patches is expressed as $\mathcal{P} = \{p_i\}_{i=1}^{H\times W}$, where $p_i = \{\boldsymbol{x}_{ij}\}_{j=1}^{M}$ and $\boldsymbol{x}_i$ is its central pixel. Each pixel $\boldsymbol{x}_{ij} \in \mathbb{R}^B$ is associated with a spectral vector and a one-hot vector $\boldsymbol{y}_{ij} \in \mathbb{R}^C$, where B and C are band number and class number, respectively. Then, spectral vectors of pixels in a patch are arranged in the feature matrix $\mathbf{X}_i \in \mathbb{R}^{M\times B}$ by row. The graph structure is represented by the adjacent matrix $\mathbf{A}_i^l \in \mathbb{R}^{M\times M}$ at the lth layer, where elements of adjacent pixels are designated as 1. $\mathbf{D}$ is the corresponding degree matrix. We transform hyperspectral patches into graph representations by treating the pixels as nodes and connecting each node to its four neighbors (left, right, top, and bottom). In SASA, $\mathbf{Q}_i^l \in \mathbb{R}^{M\times d_{\mathbf{Q}}^l}$, $\mathbf{K}_i^l \in \mathbb{R}^{M\times d_{\mathbf{K}}^l}$, and $\mathbf{V}_i^l \in \mathbb{R}^{M\times d_{\mathbf{V}}^l}$ are the query, key, and value matrix, where $d_{\mathbf{Q}}^l$, $d_{\mathbf{K}}^l$, and $d_{\mathbf{V}}^l$ are their output dimension, respectively. The hidden feature of $\mathbf{X}_i$ is $\mathbf{H}_i^l$, and especially $\mathbf{H}_i^0 = \mathbf{X}_i$. In addition, the standard deviation matrix of $\mathbf{H}_i^l$ is $\mathbf{S}_i^l \in \mathbb{R}^{M\times d_{\mathbf{V}}^l}$. For pixels, they all have M contextual views. The view_i^k is kth ($k = 1, \ldots, M$) view of pixel $\boldsymbol{x}_i$, and especially $\text{view}_i^{(M/2)} = p_i$ is its central view.

A. Data Augmentation

Although the SAGT can dynamically remove irrelevant edges in graphs, using uniform graph structures can make the model fragile and prone to learning simple patterns. Relying on a single graph structure hampers the network's ability to generalize during training and limits the diversity of feature extraction. Consequently, the trained model struggles to adapt to real-world complex scenarios.

1) Random Discard: To address this, a data augmentation technique called random discard is designed to enrich graph structures before training. Random discard works because it introduces variability and forces the model to generalize better by preventing overfitting to specific patterns in the data. In this process, each node and its corresponding edges, except for the target node, are discarded with a probability ρ. This generates various graph structures for each sample, increasing spatial diversity, providing more contextual information, and forcing the model to learn useful and discriminative features from the outset. A similar technique, dropout [55], was introduced before our work, randomly dropping intermediate neurons in the network space. However, our random discard operates in the raw data space. While dropout implicitly trains multiple subnetworks and integrates them during inference to improve generalization, our approach alters the original data structures rather than the network structures for more effective modeling and learning. Additionally, since not all nodes and edges in HSI graph construction are essential or beneficial for representation, random node discard is more appropriate and effective than dropping network neurons.

2) Random Offset: In addition to random discard, a random offset strategy is introduced to further enhance data augmentation. Random offset works by introducing positional variability into the input data, which helps the model become more robust and adaptive to changes in spatial configurations. Since we are working with graph structures, the offset range can be set within $(h/2, w/2)$ to shift the patches. This means that when extracting patch data, it is not necessary to always center on the target pixel; instead, the patch can move within the specified range. This strategy applies random displacements to the input data, simulating various real-world scenarios and helping the model better handle uncertainties caused by positional shifts.

B. Structure-Aware SA

As shown in (1), the SA in Transformers adaptively aggregates homogeneous features among pixels using a query scheme $\text{softmax}((\mathbf{Q}_i^l \mathbf{K}_i^{l\top}/(d_{\mathbf{K}})^{1/2}))$. This operation is based on content and does not account for positional information. To address this, Vaswani et al. [21] introduced 1-D positional encoding using a sine–cosine function in $\mathbf{X}_i$ to make SA sensitive to positional changes. Since image information is distributed in a 2-D space, Dosovitskiy et al. [20] replaced $\boldsymbol{\rho}_j$ with an array of learnable parameters to better represent positions. To further enhance the relative positional encoding among elements, Liu et al. [48] proposed

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a learnable relative position bias $\widehat{\mathbf{B}} \in \mathbb{R}^{(2M-1)\times(2M-1)}$, shown as

$$\mathrm{SA}\left(\mathbf{H}_i^{l-1}\right) = \mathrm{softmax}\left(\frac{\mathbf{Q}_i^l \mathbf{K}_i^{l\top}}{\sqrt{d_{\mathbf{K}}^l}} + \mathbf{B}\right)\mathbf{V}_i^l \tag{6}$$

where $\mathbf{B} \in \mathbb{R}^{M\times M}$ is taken from $\widehat{\mathbf{B}}$. Here

$$\begin{aligned}\mathbf{Q}_i^l &= \mathbf{H}_i^{l-1}\mathbf{W}_{\mathbf{Q}}^l \\ \mathbf{K}_i^l &= \mathbf{H}_i^{l-1}\mathbf{W}_{\mathbf{K}}^l \\ \mathbf{V}_i^l &= \mathbf{H}_i^{l-1}\mathbf{W}_{\mathbf{V}}^l\end{aligned} \tag{7}$$

where $\mathbf{W}_{\mathbf{Q}}^l \in \mathbb{R}^{d_{\mathbf{Q}}^{l-1}\times d_{\mathbf{Q}}^l}$, $\mathbf{W}_{\mathbf{K}}^l \in \mathbb{R}^{d_{\mathbf{K}}^{l-1}\times d_{\mathbf{K}}^l}$, and $\mathbf{W}_{\mathbf{V}}^l \in \mathbb{R}^{d_{\mathbf{V}}^{l-1}\times d_{\mathbf{V}}^l}$ are their parametric matrices, respectively. The affine transformation with respect to $\mathbf{Q}_i^l$, $\mathbf{K}_i^l$, and $\mathbf{V}_i^l$ is also applicable for SASA. Although relative position bias provides slight improvements, it requires presetting its size and is not well-suited for handling inputs of varying sizes. Therefore, a more flexible and effective positional encoding approach is desirable. We propose embedding the graph as a relative positional indicator to enhance representation and extraction. In our exploration, we find that graph structures can expand freely without repetitive construction and effectively represent relationships among objects through connected edges and associated weights. Here, the relative position information is stored in the adjacent matrix $\mathbf{A}_i^l$, where edges indicate whether objects are connected and weights represent their distances. Features of distant objects can be transmitted along connected paths, cascading cross-node relationships. To ensure numerical stability, we incorporate symmetric normalization $\mathbf{D}_i^{-1/2}\mathbf{A}_i\mathbf{D}_i^{-1/2}$ into the SA mechanism to enhance positional perception and structural representation. Our proposed SASA is illustrated as

$$\mathrm{SASA}\left(\mathbf{H}_i^{l-1}, \mathbf{A}_i^{l-1}\right) = \mathrm{softmax}\left(\frac{\mathbf{Q}_i^l \mathbf{K}_i^{lT}}{\sqrt{d_{\mathbf{K}}}} + \mathbf{D}_i^{-1/2}\mathbf{A}_i^{l-1}\mathbf{D}_i^{-1/2}\right)\mathbf{V}_i^l. \tag{8}$$

Compared to existing positional encoding methods, our graph-based indicator can flexibly expand without disrupting the original structure or adding extra training overhead.

Since the role of softmax(·) here is to normalize numerical values, we can relax this constraint to achieve more efficient computation and training. As a result, SASA can be decomposed into SA and graph convolution operators, uncovering the intrinsic spectral–spatial characteristics of HSIs from two distinct perspectives, as shown in

$$\mathrm{SASA}\left(\mathbf{H}_i^{l-1}, \mathbf{A}_i^{l-1}\right) = \mathrm{softmax}\left(\frac{\mathbf{Q}_i^l \mathbf{K}_i^{lT}}{\sqrt{d_{\mathbf{K}}}}\right)\mathbf{V}_i^l + \mathbf{D}_i^{-1/2}\mathbf{A}_i^{l-1}\mathbf{D}_i^{-1/2}\mathbf{V}_i^l. \tag{9}$$

In the above equation, the former operates on a dynamic global attention map, while the latter acts on a static local graph map. This provides a different perspective on how the graph-based positional indicator interacts with SA. Both global relationships and local details contribute to object recognition, but in different cases, the global features from attention maps and local features from graph maps may have varying importance. To address this, we design a graph-aware weighting module that adaptively determines their relative significance and fuses the features accordingly. This module uses the standard deviation $\mathbf{S}_i^l$ of graph node statistics as input. The standard deviation reflects the variability among nodes, and the result is expected to indicate the degree of graph homogeneity. The graph-aware weighting is computed as

$$\alpha^l = \mathrm{sigmoid}\left(\mathrm{MLP}\left(\mathbf{S}_i^l\right)\right) \tag{10}$$

where α^l is the output of the proposed graph-aware weighting module, derived through multilayer perceptron (MLP) transformations and nonlinear activation sigmoid(·). Finally, the structure of SASA is illustrated in Fig. 3(b), and its specific calculation is expressed as

$$\mathrm{SASA}\left(\mathbf{H}_i^{l-1}, \mathbf{A}_i^{l-1}\right) = \left(\left(1-\alpha^l\right)\mathrm{softmax}\left(\frac{\mathbf{Q}_i^l \mathbf{K}_i^{lT}}{\sqrt{d_{\mathbf{K}}}}\right) + \alpha^l \mathbf{D}_i^{-1/2}\mathbf{A}_i^{l-1}\mathbf{D}_i^{-1/2}\right)\mathbf{V}_i^l. \tag{11}$$

HSIs possess rich spectral characteristics of object components, and SASA can be viewed as a spatial attention mechanism for feature extraction. Therefore, the MLP following SA in the Transformer is replaced with a channel attention module to enhance the integration of spectral and spatial information. Additionally, all layer normalization in the Transformer is replaced with batch normalization (BN), as it provides better generalization with fewer labeled samples. For efficiency, the channel attention module takes the SASA output as input and generates a channel-attention vector, which is then multiplied with the input to emphasize important features and suppress noise, as shown in

$$\mathbf{H}_i^l = \widetilde{\mathbf{H}}_i^l * \mathrm{sigmoid}\left(\mathrm{MLP}\left(\mathrm{BN}\left(\widetilde{\mathbf{H}}_i^l\right)\right)\right). \tag{12}$$

C. Structure-Adaptive Pooling

Datasets in the graph domain [56] have explicit graph structures, whereas in HSIs, the structure is not explicit. Handcrafted graphs can interfere with homogeneous information extraction in SASA due to design noise and limited capacity for structural optimization. Therefore, it is necessary to adjust the graph structure dynamically during SA feature extraction. We propose a novel SAP based on cosine similarity, which progressively discards edges with low confidence to make the graphs more homogeneous and accurate. This operation helps adaptively optimize the graph structure, improving positional information and making the model less sensitive to hand-crafted noise. Specifically, a hidden feature $\mathbf{H}_i^l$ of nodes is passed through the MLP and transformed into $\mathbf{Z}_i^l$ to remove irrelevant factors. This is then multiplied by its transpose to generate a similarity matrix for edge discard evaluation. The similarity matrix is calculated dynamically and trained iteratively to adjust the graph structure. Ultimately, a quantitative matrix derived from the function $\mathrm{sign}_\beta(\cdot)$ serves as a mask to remove edges with confidence lower than β. Notably, the retained values replace corresponding ones in $\mathbf{A}_i^l$ as edge weights in subsequent layers. This allows gradients to flowthrough the pooling operation, updating parameters

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during backpropagation and facilitating feature abstraction and network convergence. The SAP is computed as

$$\mathbf{Z}_i^l = \mathrm{MLP}\left(\mathbf{H}_i^l\right) \tag{13}$$

$$\mathbf{A}_i^l = \mathrm{sign}_\beta\left(\mathbf{Z}_i^l \mathbf{Z}_i^{l^T}\right) \odot \mathbf{A}_i^{l-1} \tag{14}$$

$$\mathrm{sign}_\beta(x) = \begin{cases} 1, & x \geq \beta \\ 0, & x < \beta \end{cases} \tag{15}$$

where β denotes the confidence threshold to control edge retention and discard and x is the cosine similarity of paired nodes. The threshold β for edge filtering in graph pooling is empirically set to 0, as the cosine similarity falls within the range of $[-1, 1]$.

D. Structure-Adaptive Graph Transformer

We integrate SASA and SAP into an effective framework to form the SAGT. The SAGT enhances the extraction of homogeneous information by incorporating a more robust positional indicator and a more accurate graph structure, mitigating the effects of spectral variability and boundary complexity in HSIs, as illustrated in Fig. 3(a). The SAGT consists of $\boldsymbol{N}$ graph Transformer blocks and a SASA module. While a single graph Transformer block can handle relationships between close neighbors, stacking multiple blocks allows information to flow across distant nodes, improving the model's capacity to understand the global structure. Each graph Transformer block includes an SAP module, a SASA module, and a skip connection operation for cascading feature representation and structure optimization. A SASA module is placed at the first layer to transform the initial representation. The SAGT is trained end-to-end for HSI graph construction, node representation, feature extraction, and category recognition.

E. Multiview Decision

During data preprocessing, the HSI is divided into overlapping patches, where each pixel appears in multiple patches and is surrounded by different contexts, or views of objects. For each pixel, some views provide rich content for recognition, while others contain less relevant information. Generally, the farther a view is from the target pixel, the less homogeneous information it contains, with the most important data residing in the central view. Therefore, it is crucial to enhance feature extraction during training and improve object recognition during prediction. The central views, $\mathrm{view}_i^{(h\times w/2)}$, are used to effectively train the network and learn features during training, while other relevant views are employed for comprehensive category prediction during testing. We propose a multiview decision approach during postprocessing, integrating information from various views of a pixel and determining the final prediction through majority voting, which is shown in Fig. 4. This multiview decision aims to enhance classification by leveraging diverse contextual perspectives in postprocessing, described as

$$\mathrm{pred}_i = \mathrm{argmax} \sum_{k=1}^{h\times w} \mathcal{F}\left(\mathrm{view}_i^k\right) \tag{16}$$

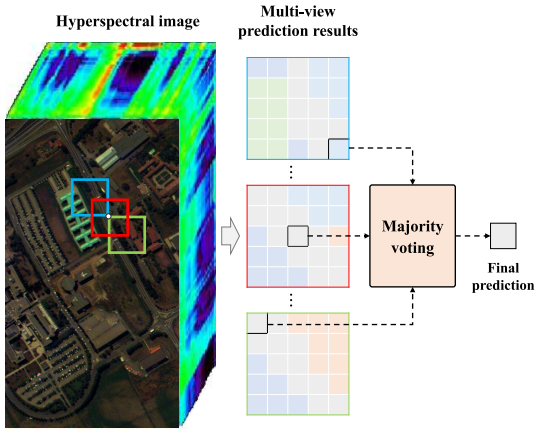


Fig. 4. Pipeline of the multiview decision-making strategy. During inference, it integrates predictions of each pixel across different views using majority voting, improving the robustness of HSI classification.

where $\mathcal{F}$ represents our classification model.

On the one hand, multiview decision integrates information from multiple contexts and strikes a balance between efficiency and performance, even though the clipping operation may disrupt the complete distribution of objects. On the other hand, multiview decision takes into account predictions from various contextual views, making our model more resilient to noise and more robust in complex scenarios. If one of the views is noisy or contains misleading contextual information (e.g., the pixel on the boundary between classes), the multiview approach mitigates the impact of that noise by considering multiple perspectives. Consequently, the SAGT can more effectively capture the underlying graph structure in HSIs and learn intrinsic spectral–spatial semantics. Additionally, data augmentation and multiview decision further enhance the model's generalization and robustness.

IV. Experiments

In this section, a series of ablation and comparison experiments is conducted on benchmark datasets to evaluate the performance of our proposed method. The ablation experiments explore the contribution of each module in the architecture, while the comparison experiments compare SAGT with other state-of-the-art models, demonstrating its effectiveness for HSI classification.

A. Experimental Setups

In data preprocessing, the training set consists of ten randomly selected samples from each class, while the remaining samples form the testing set. Since the choice of training samples can impact the model's performance, particularly when there are few labeled samples, the selection process is repeated ten times randomly for both training and testing. Across all benchmark datasets, we empirically crop HSIs into overlapping 11×11 patches. The classification performance is evaluated using overall accuracy (OA), average accuracy (AA), and the kappa coefficient, with the mean values recorded to ensure a fair evaluation.

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TABLE I
LAND-COVER CLASSES WITH NUMBER OF SAMPLES PER CLASS FOR THE PAVIAU DATASET

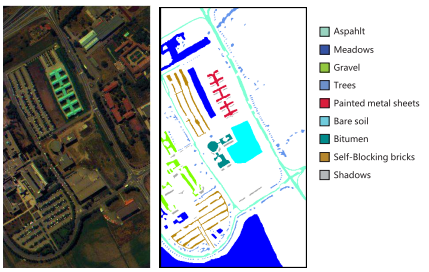
NO.	Class	Train	Test
C1	Asphalt	10	6,621
C2	Meadows	10	18,639
C3	Gravel	10	2,089
C4	Trees	10	3,054
C5	Painted metal sheets	10	1,335
C6	Bare Soil	10	5,019
C7	Bitumen	10	1,320
C8	Self-Blocking Bricks	10	3,672
C9	Shadows	10	937
	Total	90	42,686

Additionally, several hyperparameters need to be appropriately configured before running the model. First, each SASA module contains 128 neurons and a single head for homogeneous feature extraction, balancing network capability and computational efficiency. The threshold β for edge filtering in graph pooling is set to 0, as the cosine similarity falls within the range of $[-1, 1]$. Second, the overall SAGT model is constructed with one SASA module and one graph Transformer block. Since the SASA enables rapid acquisition and aggregation of global context within the graph, a shallow model with fewer parameters can capture sufficient structural and feature-related information, obtaining good performance for HSI classification. A deeper network with multiple graph Transformer blocks may lead to overfitting, particularly when there is a limited amount of labeled data. Empirically, a single graph Transformer block can strike a balance between model simplicity and performance, integrated with an attention mechanism of SASA that allows for efficient information flow between nodes. During the training phase, the Adam optimizer with an initial learning rate of 0.001 is used to automatically update the learning parameters, and a mini-batch size of 30 is used for 300 iterative epochs.

B. Description of HSI Datasets

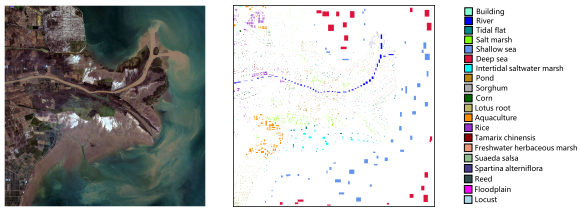
In the experiment, we select three HSIs from the Pavia University (PaviaU), Yellow River Estuary (YRE), and Yan Cheng (YC) as benchmark datasets. These images are captured at different altitudes by various hyperspectral sensors, containing multiple ground objects across diverse scenarios. These varied characteristics are advantageous for validating the effectiveness and robustness of the SAGT model. They are presented as follows.

1) *PaviaU*: This scene was captured by the reflective optics spectrographic imaging system (ROSIS) sensor during a flight over PaviaU, Italy, with a spatial resolution of 1.3 m/pixel. The image size is 610 × 340, with 42 776 pixels labeled across nine categories. Initially, the PaviaU image contains 115 spectral bands in the range of 0.43–0.86 μm. After removing 12 noisy bands, the resulting HSI includes 103 bands. The false-color image and ground-truth map are shown in Fig. 5, and the sample numbers are listed in Table I.
2) *YRE*: The YRE dataset was collected by the visible shortwave infrared advanced hyperspectral imager (AHSI) sensor mounted on China's Gaofen-5 satellite, with a spatial resolution of 30 m/pixel. It contains 330 spectral bands ranging from 0.4 to 2.5 μm, including 150 visible and near-infrared (VNIR) bands and 180 shortwave infrared (SWIR) bands. The YRE image has a size of 1400 × 1400, with 77 937 pixels labeled across 20 categories. The false-color image and ground-truth map are shown in Fig. 6, and the sample numbers are listed in Table II.
3) *YC*: The YC scene was captured by the AHSI sensor over a local area in Yancheng, Jiangsu province, with 266 spectral bands remaining after removing the noisy ones. The image size is 585 × 1175, with 7112 pixels labeled across 20 categories. The false-color image and



(a) (b)

Fig. 5. (a) False-color image and (b) ground-truth map of the PaviaU dataset.



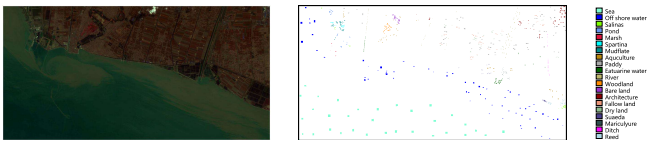
(a) (b)

Fig. 6. (a) False-color image and (b) ground-truth map of the YRE dataset.

TABLE II
LAND-COVER CLASSES WITH NUMBER OF SAMPLES PER CLASS FOR THE YRE DATASET

No.	Class	Train	Test
C1	Building	10	523
C2	River	10	5,366
C3	Salt marsh	10	4,985
C4	Shallow sea	10	17,540
C5	Deep sea	10	18,667
C6	Intertidal saltwater marsh	10	2,333
C7	Tidal flat	10	1,782
C8	Pond	10	1,777
C9	Sorghum	10	636
C10	Corn	10	1,499
C11	Lotus root	10	2,709
C12	Aquaculture	10	8,009
C13	Rice	10	5,498
C14	Tamarix chinensis	10	1,210
C15	Freshwater herbaceous marsh	10	1,407
C16	Suaeda salsa	10	864
C17	Spartina alterniflora	10	570
C18	Reed	10	1,960
C19	Floodplain	10	337
C20	Locust	10	65
	Total	200	77,737

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(a) (b)

Fig. 7. (a) False-color image and (b) ground-truth map of the YC dataset.

TABLE III
LAND-COVER CLASSES WITH NUMBER OF SAMPLES PER CLASS FOR THE YC DATASET

No.	Class	Train	Test
C1	Sea	10	2,385
C2	Off shore water	10	1,295
C3	Salina	10	100
C4	Pond	10	123
C5	Marsh	10	11
C6	Spartina	10	73
C7	Mudflate	10	208
C8	Aquculture	10	218
C9	Paddy	10	822
C10	Eatuarine water	10	162
C11	River	10	207
C12	Woodland	10	146
C13	Bare land	10	144
C14	Architecture	10	350
C15	Fallow land	10	224
C16	Dry land	10	194
C17	Suaeda	10	75
C18	Mariculyure	10	25
C19	Ditch	10	28
C20	Reed	10	122
	Total	200	6,912

ground-truth map are shown in Fig. 7, and the sample numbers are listed in Table III.

C. Ablation Experiments

In this study, a series of ablation experiments about the proposed modules is conducted to explore their effectiveness in the model.

1) Structure-Aware SA: The effectiveness of SASA is validated from two aspects. On the one hand, we compare the SASA with other common positional encoding approaches, including the sine–cosine function, learnable parameters, and relative position bias. On the other hand, from the perspective of information integration, SASA can be viewed as a coupling of the Transformer and GCN, so we also compare it with a Transformer without positional encoding and a GCN. Table IV presents the comparative results of the Transformer with multiple positional encodings, GCN, and our SASA module, termed SAGT w/o pooling. The Transformer used here consists of two encoder layers without a decoder, while the GCN has two corresponding convolutional layers. SAGT w/o pooling also includes two SASA modules but without SAP.

It is evident that the Transformer with the other three positional encodings performs worse than our SASA in terms of accuracy. The sine–cosine encoding cannot effectively represent 2-D positions in HSIs, and both learnable approaches increase the training burden and requirements. In contrast, our SASA does not introduce additional parameters and embeds a robust graph-based encoding to flexibly represent relative positions. As shown in Fig. 8, when we progressively increase

TABLE IV
EXPERIMENTAL RESULTS OF DIFFERENT BACKBONES ON THREE HSI DATASETS

Dataset	Model	Position	OA(%)	AA(%)	kappa	param.
PaviaU	Transformer	✗	83.68	84.09	0.79	33K
		sin-cos	80.63	82.35	0.75	33K
		learnable	82.35	82.72	0.77	36K
		relative	83.19	83.34	0.78	34K
	GCN	✗	82.80	84.96	0.78	33K
	SAGT w/o pooling	✗	**86.81**	**88.66**	**0.83**	33K
YRE	Transformer	✗	87.97	81.12	0.86	68K
		sin-cos	86.47	79.71	0.84	68K
		learnable	84.82	75.89	0.82	75K
		relative	87.14	81.22	0.85	69K
	GCN	✗	89.04	84.81	0.87	68K
	SAGT w/o pooling	✗	**89.58**	**85.50**	**0.88**	68K
YC	Transformer	✗	93.02	90.51	0.92	65K
		sin-cos	92.25	89.63	0.91	65K
		learnable	90.64	88.05	0.89	72K
		relative	93.50	90.15	0.92	66K
	GCN	✗	95.23	94.33	0.94	65K
	SAGT w/o pooling	✗	**96.02**	**94.39**	**0.95**	65K

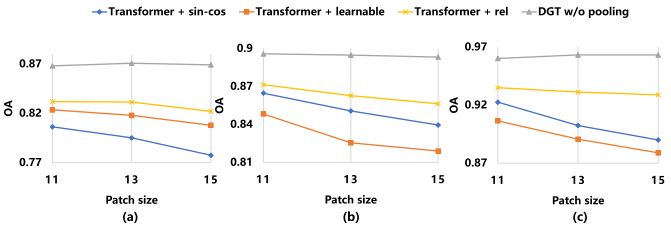


Fig. 8. Change in OA of different positional encodings as the length of input sequences increases. All models are trained with a patch size of 11 × 11 and evaluated at patch sizes of 11 × 11, 13 × 13, and 15 × 15, respectively. (a) PaviaU. (b) YRE. (c) YC.

the input size during inference, there is a noticeable decrease in OA for the Transformer with sine–cosine and learnable encodings across three datasets, while the Transformer with relative position bias shows a slight decline. This demonstrates that relative positional encoding is more robust than absolute encoding, but it still experiences a drop in OA when the patch size is increased to 15. The need for interpolation over a broader range to handle longer input blurs positional distinctions and distorts the original distribution. In contrast, our SASA maintains stable performance and high accuracy across varying input sizes, as the graph-based positional encoding is scalable to preserve the positional information internally.

From another perspective, our SASA can be seen as an integration of local details and global relationships. As shown in Table IV, SAGT w/o pooling outperforms both the Transformer and GCN on benchmark datasets in terms of OA, AA, and kappa. On the PaviaU dataset, the Transformer achieves better performance than GCN, likely because the high spatial resolution limits contextual perception within the same neighborhood range, making the Transformer more effective for rapid aggregation of global features. In contrast, for HSIs with lower spatial resolution, such as YRE and YC, GCN excels by focusing on local topologies and structural components for detailed extraction. This suggests that the spatial resolution of HSIs may influence model performance. Moreover, our SASA integrates the local and global advantages, avoiding the limitations of either approach. By dynamically incorporating information from both Transformer and GCN through graph-aware weighting, it adapts to different datasets effectively.

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TABLE V

EXPERIMENTAL RESULTS OF SAGT WITH AND WITHOUT THE CNN BRANCH, RANDOM DISCARD, SAP, RANDOM OFFSET, AND MULTI-VIEW DECISION ON THREE HSI DATASETS

Dataset	CNN branch	Random discard	SAP	Multi-view	Random offset	OA(%)	AA(%)	kappa
PaviaU	✗	✗	✗	✗	✗	85.69	88.92	0.81
	✓	✗	✗	✗	✗	86.81	88.66	0.83
	✓	✗	✓	✗	✗	87.88	89.27	0.84
	✓	✓	✓	✗	✗	91.63	91.64	0.89
	✓	✓	✓	✓	✗	93.28	93.41	0.91
	✓	✓	✓	✓	✓	**94.23**	**95.65**	**0.92**
YRE	✗	✗	✗	✗	✗	88.77	83.45	0.87
	✓	✗	✗	✗	✗	89.58	85.50	0.88
	✓	✗	✓	✗	✗	90.76	86.15	0.89
	✓	✓	✓	✗	✗	91.54	87.72	0.90
	✓	✓	✓	✓	✗	91.87	88.31	0.91
	✓	✓	✓	✓	✓	**92.79**	**89.23**	**0.92**
YC	✗	✗	✗	✗	✗	95.28	94.36	0.94
	✓	✗	✗	✗	✗	96.02	94.39	0.95
	✓	✗	✓	✗	✗	97.78	96.72	0.97
	✓	✓	✓	✗	✗	99.12	98.38	0.99
	✓	✓	✓	✓	✗	99.19	98.55	0.99
	✓	✓	✓	✓	✓	**99.28**	**98.94**	**0.99**

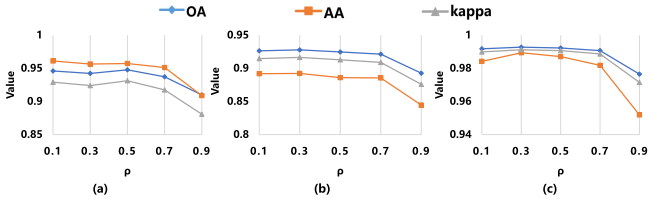


Fig. 9. Performance of SAGT with different values of ρ on three HSI datasets. (a) PaviaU. (b) YRE. (c) YC.

2) CNN Branch: The experimental results of SAGT with and without the CNN branch are reported in the first and second rows of Table V, showing that the inclusion of the CNN branch consistently improves classification performance. For instance, on PaviaU, the OA increases from 85.69% to 86.81%, and similar gains are observed on YRE (from 88.77% to 89.58%) and YC (from 95.28% to 96.02%). These results indicate that exploring spatial details, such as local textures, in HSI patches through a CNN branch can enhance the semantic discriminability of node representations and thereby improve classification accuracy. After adding the graph Transformer to the CNN, the last row achieves the best accuracy. This finding also highlights the complementary advantages of CNN and graph Transformer, where the former focuses on spatial details while the latter aggregates homogeneous features.

3) Data Augmentation: First, the comparison results of SAGT with and without data augmentation through random discard are quantitatively presented in the third and fourth rows for each dataset in Table V. Overall, the model shows improved OA, AA, and kappa with data augmentation, as the number and diversity of training samples are increased. In certain localized areas, discarding some heterogeneous nodes enriches the graph contexts and provides more informative samples. The influence of increasing discard probability ρ on OA, AA, and kappa, with intervals of 0.2 from 0.1 to 0.9, is shown in Fig. 9. In general, a larger ρ means that more nodes are discarded during data preprocessing, leading to sparser node connections and more diverse graph structures. However,

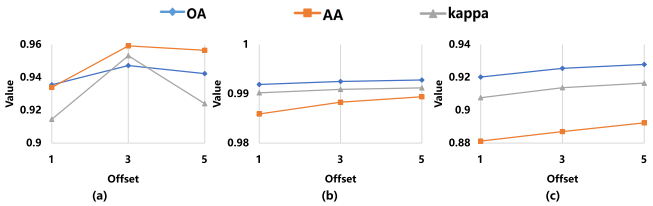


Fig. 10. Performance of SAGT with different offsets on three HSI datasets. (a) PaviaU. (b) YRE. (c) YC.

if ρ becomes too large, it can excessively reduce the amount of available information and degrade model performance. The experimental results indicate that gradually increasing ρ initially improves accuracy until it reaches a peak at a certain value, after which performance declines. There is a sharp drop in performance when ρ increases from 0.7 to 0.9, resulting from more incomplete graph structures and less useful information. Based on a comprehensive evaluation, ρ is set to 0.3 for three HSI datasets to achieve stable accuracy.

Second, the comparison results of SAGT with and without data augmentation through random offset are presented in the fifth and sixth rows in Table V. Clearly, applying the random offset technique during training improves SAGT's performance in terms of OA, AA, and kappa. Random offset encourages the model to focus more on graph structures rather than just central node features, enhancing contextual understanding and improving classification. The positive effect of random offset is more significant on the PaviaU and YRE datasets than on YC, likely due to the greater complexity of the spatial content in those two datasets. Fig. 10 illustrates the impact of increasing offset values from 1 to 5, in intervals of 2, on OA, AA, and kappa. A smaller offset may have a limited effect on contextual information, while a larger offset introduces more positional variation. The experimental results indicate that increasing the offset value improves performance. Nevertheless, excessive offsets may lead to asymmetric contexts for label nodes. The optimal offset for PaviaU is 3, while for YRE and YC, it is 5. To ensure robustness in practice, we set the offset value to 5 for all datasets.

4) Structure-Adaptive Pooling: The experimental results of SAGT with and without SAP are reported in the second and third rows of Table V, showing that SAP enhances SAGT performance across the three benchmark datasets. Hand-crafted noise and mixed pixels reduce the discriminability of ground objects and hinder the aggregation of homogeneous features in SASA. Fortunately, SAP dynamically adjusts graphs during propagation to better capture positional information, allowing SAGT to tolerate these interfering factors. It guides the model in learning better feature hierarchies, making it more robust to variations in data quality and spatial complexity. Additionally, SAP creates a coarse-to-fine structure that guides feature aggregation, promoting iterative graph optimization and cascading feature extraction. This approach ensures that the model captures both global context and local details, making the feature extraction process more effective and leading to better classification accuracy.

5) Multiview Decision: In postprocessing, an ensemble learning strategy of multiview decision is applied after SAGT

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TABLE VI

CLASSIFICATION ACCURACY (%) AND KAPPA COEFFICIENT FOR THE PAVIAU DATASET WITH VARIOUS ALTERNATIVES. RESULTS OF TWO-CNN, 3DVSCNN, HSGAN, SSLSTMS, 3DCAE, MDL4OW, AND GIGCN ARE FROM PAPER [47], WHILE OTHERS ARE REIMPLEMENTED BY US

class	Two-CNN	3DVSCNN	HSGAN	SSLSTMs	SSCL	3DCAE	MDL4OW	SAGEConv	Spectralformer	SSFTT	DBCTNet	GiGCN	SAGT
1	72.38±5.55	66.52±8.92	68.91±7.09	72.59±7.75	51.20±5.29	49.41±25.25	72.77±11.92	81.22±5.90	47.28±16.52	80.72±8.35	90.55±3.44	**95.28±5.25**	94.52±4.66
2	87.47±6.12	84.04±4.31	77.77±10.61	68.86±16.69	77.02±9.17	51.60±24.70	76.24±9.21	68.84±8.02	86.36±5.92	**95.67±1.67**	92.03±6.03	91.61±6.54	94.24±4.08
3	48.44±8.04	75.44±11.04	39.01±22.37	48.08±10.70	71.53±10.31	54.06±17.95	70.39±5.60	53.82±19.13	68.53±11.60	68.74±10.79	86.06±8.89	**99.27±2.87**	94.16±8.49
4	96.50±2.37	81.95±9.90	81.64±7.80	79.06±10.95	60.51±11.50	94.81±4.14	93.21±2.98	88.18±13.14	84.18±2.85	93.01±3.84	92.00±5.37	83.41±2.36	**98.51±2.42**
5	95.41±5.00	99.73±0.44	98.46±1.70	93.80±2.76	87.75±9.07	99.86±0.19	99.41±0.46	99.02±2.84	99.85±0.03	96.50±3.98	99.48±0.53	**99.99±0.36**	99.96±0.03
6	56.50±8.74	69.11±14.85	43.00±13.39	62.53±14.25	53.97±6.30	57.40±27.87	71.44±13.97	67.48±12.02	35.86±13.10	81.05±10.26	**98.63±0.87**	97.84±3.57	87.07±2.68
7	62.66±8.59	69.33±20.51	83.99±10.87	65.39±14.82	54.02±17.15	80.34±27.24	88.03±8.43	83.38±3.55	83.24±7.70	79.01±9.55	98.52±9.03	**99.92±0.03**	97.25±0.73
8	67.47±6.36	55.45±5.24	79.79±9.75	67.60±15.38	61.58±11.38	57.97±22.02	63.80±12.85	76.90±21.25	56.21±7.69	72.17±10.39	84.44±4.07	**97.48±3.62**	95.26±7.52
9	97.35±3.90	77.65±10.40	99.33±1.32	97.02±5.84	72.91±9.39	98.76±1.27	97.47±1.67	99.12±0.54	90.83±2.57	96.23±4.73	95.19±6.53	81.84±3.95	**99.90±0.09**
AA(%)	76.02±1.33	75.47±4.23	74.66±2.40	72.77±4.78	65.61±4.42	71.58±2.83	81.42±2.56	79.77±3.37	72.48±3.20	84.79±3.64	92.99±1.77	94.07±1.63	**95.65±0.84**
OA(%)	78.20±1.88	76.43±2.76	72.17±2.06	69.59±6.16	67.06±4.25	59.14±6.69	76.55±4.97	74.00±4.61	71.16±4.54	87.63±2.98	92.13±2.47	93.51±3.35	**94.23±2.48**
kappa	0.71±0.022	0.69±0.036	0.64±0.027	0.62±0.066	0.58±0.047	0.51±0.058	0.70±0.056	0.67±0.051	0.62±0.056	0.83±0.040	0.89±0.031	**0.92±0.043**	**0.92±0.031**

TABLE VII

CLASSIFICATION ACCURACY (%) AND KAPPA COEFFICIENT FOR THE YRE DATASET WITH VARIOUS ALTERNATIVES. RESULTS OF TWO-CNN, 3DVSCNN, HSGAN, SSLSTMS, 3DCAE, MDL4OW, AND GIGCN ARE FROM PAPER [47], WHILE OTHERS ARE REIMPLEMENTED BY US

class	Two-CNN	3DVSCNN	HSGAN	SSLSTMs	SSCL	3DCAE	MDL4OW	SAGEConv	Spectralformer	SSFTT	DBCTNet	GiGCN	SAGT
1	73.50±9.75	82.33±5.10	66.20±7.93	35.54±12.44	61.93±7.33	31.53±6.80	75.91±6.43	74.93±8.74	46.10±13.80	82.55±20.93	84.61±9.15	85.01±9.76	**88.18±3.15**
2	98.78±1.94	95.61±5.49	98.19±2.86	94.98±5.54	94.22±4.99	**99.93±0.04**	99.25±0.66	98.64±2.16	97.46±4.88	95.96±3.03	87.89±1.29	98.26±1.68	99.61±1.78
3	75.09±7.67	74.77±11.49	89.63±3.36	74.92±8.18	53.63±12.41	69.85±5.84	81.32±5.16	87.42±6.00	78.12±3.57	89.63±7.13	71.83±5.66	89.84±3.73	**90.41±2.61**
4	92.18±4.38	88.07±4.96	92.84±2.88	76.62±10.42	84.33±4.00	50.67±13.09	89.71±4.13	94.09±3.47	91.24±0.93	93.28±2.49	88.58±1.98	**98.27±2.04**	96.38±3.29
5	98.08±1.26	96.20±3.00	97.88±1.34	82.00±13.06	94.73±5.14	**100.00±0.00**	97.55±1.80	98.41±1.19	97.41±0.73	96.71±2.49	97.12±1.98	98.28±2.06	98.24±3.29
6	75.02±9.16	69.61±9.89	71.21±9.36	65.22±7.89	78.20±7.94	75.63±7.09	81.53±7.18	71.89±5.36	79.00±4.80	**82.84±12.86**	76.39±10.14	81.95±10.22	72.27±8.23
7	64.87±4.39	61.31±7.19	63.87±6.73	53.75±8.11	57.26±9.06	60.17±17.21	64.39±3.82	49.15±10.00	62.31±3.86	62.38±6.64	65.01±5.68	68.23±6.08	**72.41±5.18**
8	84.63±5.11	88.06±5.16	89.81±2.66	46.13±26.57	65.74±8.07	53.47±14.49	82.88±9.12	86.12±3.64	83.97±13.19	78.28±7.56	**95.18±16.46**	91.96±6.91	91.15±4.73
9	83.13±8.20	85.38±13.80	78.16±9.36	47.30±6.53	77.08±6.54	45.75±7.09	87.23±7.18	80.00±5.36	63.74±4.86	87.46±12.86	**93.34±7.56**	90.44±16.46	90.55±6.91
10	77.25±9.48	79.63±11.35	74.81±9.87	52.67±11.10	62.04±11.64	74.04±19.26	79.95±9.12	64.16±7.73	69.83±7.61	87.07±7.63	**91.12±7.51**	81.91±6.80	80.58±6.99
11	86.78±12.52	80.04±15.11	71.13±11.27	50.39±11.10	77.06±16.80	43.36±9.12	73.57±7.78	64.64±7.73	63.80±12.07	88.08±7.31	86.06±7.51	**91.04±6.80**	88.93±6.99
12	74.12±12.52	79.32±15.11	84.24±11.27	52.03±11.10	74.71±16.80	69.52±19.26	78.63±9.12	82.10±7.73	80.52±12.07	81.93±7.31	87.72±7.51	**90.46±6.80**	88.97±6.99
13	77.16±6.54	79.21±12.52	81.03±9.14	63.92±10.86	68.32±8.45	54.05±12.09	81.64±10.05	**90.33±7.73**	76.40±7.63	85.43±9.39	88.65±6.11	87.28±12.07	89.15±10.61
14	81.31±12.52	77.63±15.11	87.26±11.27	42.46±11.10	58.49±16.80	54.55±19.26	85.45±9.12	86.75±7.73	77.56±12.07	61.39±7.31	70.11±11.22	88.61±6.80	**91.87±2.67**
15	92.75±4.39	66.86±10.30	91.12±8.77	61.37±12.59	57.51±12.07	91.39±9.94	85.69±3.86	**95.43±6.64**	76.65±5.68	77.76±6.08	93.93±5.18	90.60±12.52	95.13±16.94
16	92.01±7.76	82.25±5.19	94.78±6.14	65.39±10.77	76.83±12.73	71.69±22.62	92.85±5.84	89.70±3.16	79.34±4.07	91.07±3.37	66.01±12.19	84.80±3.14	**96.12±5.19**
17	91.68±10.06	73.72±17.07	85.16±12.86	46.05±3.17	92.39±12.19	36.77±8.34	90.67±5.08	80.63±10.13	66.77±7.59	83.27±9.89	**100.00±0.00**	97.14±3.96	96.96±8.10
18	62.97±7.94	60.91±10.13	68.83±7.39	51.22±8.45	40.14±19.61	24.52±9.65	67.67±9.83	60.37±6.61	52.06±7.27	61.77±6.16	53.19±9.89	58.04±6.16	**79.96±3.23**
19	77.12±9.87	63.80±11.35	72.61±9.68	59.14±12.70	71.72±10.11	25.73±12.09	79.88±10.60	67.12±11.08	59.05±12.59	66.28±9.93	84.72±19.26	66.94±14.68	**88.84±7.79**
20	88.92±12.52	95.69±15.11	71.23±11.27	49.85±10.86	75.54±8.45	86.92±12.09	92.46±10.05	61.38±7.73	71.38±7.63	81.02±4.45	**100.00±0.00**	91.23±9.39	88.92±8.10
AA(%)	82.37±0.99	79.02±2.84	81.50±1.42	58.55±7.30	71.09±2.50	60.98±1.67	83.41±1.54	79.16±1.74	73.64±3.93	81.07±2.59	84.07±1.25	86.52±1.83	**89.23±1.39**
OA(%)	87.08±0.87	84.99±2.01	88.84±1.03	70.29±7.16	79.21±2.68	70.58±3.42	87.46±1.04	88.55±0.90	85.13±1.45	87.91±1.53	87.38±0.94	92.50±0.98	**92.79±0.48**
kappa	0.85±0.010	0.83±0.023	0.87±0.012	0.66±0.083	0.76±0.030	0.66±0.041	0.86±0.012	0.87±0.010	0.83±0.017	0.86±0.018	0.85±0.011	0.91±0.011	**0.92±0.006**

to integrate information from different contextual views for making a final decision. The experimental results of SAGT with and without multiview decision are shown in the fourth and fifth rows of Table V. SAGT with multiview decision achieves further improvements in classification accuracy and consistency, as integrating predictions from different views compensates for information loss and connects multiple relationships. Apart from the PaviaU dataset, the effect of multiview decision is less significant on YRE and YC, likely due to their lower spatial resolution and limited neighborhood size. At low spatial resolution, each ground object corresponds to fewer pixels, so multiple perspectives may offer similar contextual information or even introduce irrelevant surroundings beyond the central view. Overall, multiview decision enhances feature diversity and information integrity, particularly benefiting HSI classification at high spatial resolution.

D. Comparative Experiments

Comparative experiments are conducted on SAGT and state-of-the-art models in terms of quantitative and qualitative analyses, namely Two-CNN [57], 3DVSCNN [8], HSGAN [58], SSLSTMs [59], SSCL [60], 3DCAE [7], MDL4OW [61], SAGEConv [38], SpectralFormer [34], SSFTT [26], DBCTNet [62], and GiGCN [47]. These methods encompass various architectures (e.g., CNN, RNN, GCN, Transformer, etc.) and different learning paradigms (e.g., transfer learning, active learning, few-shot learning, etc.). We follow the same dataset partitioning and training configuration as GiGCN, so the results of Two-CNN, 3DVSCNN, HSGAN, SSLSTMs, 3DCAE, MDL4OW, and GiGCN are from paper [47], while others are reimplemented by us.

The quantitative experimental results on benchmark datasets are shown in Tables VI–VIII, with the highest performance for each class highlighted in bold. Overall, SAGT significantly outperforms other state-of-the-art models on the benchmark datasets. It consistently achieves better recognition in terms of OA, AA, and kappa, demonstrating that the HSI classification results produced by our model are reliable and comprehensive, rather than locally optimal. Two-CNN transfers knowledge from a source domain to another target domain to reduce the reliance on large labeled datasets, while 3DVSCNN employs an active learning strategy to select valuable samples from existing labeled ones. Both 3DCAE and MDL4OW are based on autoencoder frameworks, with the latter designed for end-to-end training and prediction. HSGAN generates additional false samples to improve feature discriminability and enhance classifier judgment. Although these learning paradigms help improve generalization under small sample conditions, they still fall short of our SAGT model due to simpler data augmentation techniques and insufficient network capabilities. Compared to SAGEConv and GiGCN, SAGT dynamically adjusts graph structures during iterative propagation, enhancing intraclass similarity and interclass distinction for feature representation. While

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TABLE VIII

CLASSIFICATION ACCURACY (%) AND KAPPA COEFFICIENT FOR THE YC DATASET WITH VARIOUS ALTERNATIVES. RESULTS OF TWO-CNN, 3DVSCNN, HSGAN, SSLSTMS, 3DCAE, MDL4OW, AND GIGCN ARE FROM PAPER [47], WHILE OTHERS ARE REIMPLEMENTED BY US

class	Two-CNN	3DVSCNN	HSGAN	SSLSTMs	SSCL	3DCAE	MDL4OW	SAGEConv	Spectralformer	SSFTT	DBCTNet	GiGCN	SAGT
1	90.51±4.17	76.12±6.73	97.10±1.98	97.85±1.70	91.05±3.57	44.53±43.55	90.53±4.67	98.30±0.67	93.38±3.89	99.81±0.73	99.49±1.88	**100.00±0.00**	99.95±0.04
2	97.42±1.30	96.59±5.19	**99.93±0.11**	99.15±0.93	97.38±2.24	99.19±1.34	99.13±0.77	99.85±0.46	95.10±1.60	99.79±0.54	99.56±0.63	99.50±0.19	99.91±0.19
3	81.70±7.73	77.70±12.06	83.80±12.78	62.10±24.73	70.30±8.92	71.70±11.40	85.90±9.71	84.50±14.91	54.00±7.81	89.60±7.32	89.20±6.85	91.20±3.88	**96.20±5.39**
4	70.16±6.14	66.50±6.43	83.90±6.93	75.45±7.17	59.02±12.88	66.42±13.75	96.10±5.67	88.78±11.87	65.04±7.45	90.08±11.28	96.59±6.81	98.46±0.87	**98.78±0.65**
5	73.64±10.00	92.73±7.17	75.45±23.49	71.82±16.29	78.18±23.16	94.55±11.50	**100.00±0.00**	85.45±22.11	77.27±13.33	92.72±36.52	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
6	85.21±5.66	90.96±13.88	97.81±3.83	72.88±19.93	77.40±11.61	**100.00±0.00**	99.32±1.16	**100.00±0.00**	77.67±8.94	96.57±6.83	99.73±4.91	98.77±1.08	99.73±0.08
7	62.60±9.00	75.53±9.69	89.90±6.93	71.20±10.75	60.62±13.06	62.16±31.40	93.80±5.01	89.57±5.15	62.93±14.89	95.86±4.17	94.33±4.45	87.45±1.00	**97.88±2.63**
8	71.24±8.01	73.17±12.68	93.17±5.98	77.25±10.12	54.31±12.19	83.85±12.83	96.97±1.84	93.35±5.35	67.06±10.76	96.83±2.10	99.63±2.49	99.63±0	**99.95±0.62**
9	79.51±7.03	80.63±10.67	94.76±3.87	93.73±4.78	60.55±11.10	96.90±0.88	96.92±1.19	97.10±0.44	81.73±11.34	96.98±0.97	98.27±1.08	96.84±0.89	**99.78±0.06**
10	71.91±11.77	87.04±6.89	90.25±4.60	80.68±7.53	70.12±11.24	81.23±1.85	97.04±7.04	90.12±3.28	75.25±10.42	96.35±2.80	97.41±1.36	97.35±0.98	**98.40±1.60**
11	91.50±6.83	47.15±13.78	69.95±10.21	91.69±4.86	78.50±11.05	89.42±3.52	**97.78±1.29**	85.41±2.30	96.33±5.57	96.85±1.83	94.15±6.89	90.68±4.24	97.54±1.71
12	94.32±5.61	68.01±40.64	99.25±1.56	93.36±4.78	72.88±13.69	99.59±0.35	99.93±0.22	99.32±1.05	86.92±9.90	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
13	82.99±9.49	89.31±6.52	92.22±9.01	70.97±15.74	82.15±4.70	84.24±9.60	97.08±2.47	97.99±3.07	54.86±4.64	97.15±13.48	99.79±0.21	99.58±0.32	**100.00±0.28**
14	52.71±10.18	45.00±13.89	52.83±16.33	41.31±9.94	48.71±5.88	6.63±3.43	76.66±8.30	80.74±11.15	25.89±7.19	87.02±6.78	**95.54±4.28**	92.37±5.93	93.29±1.13
15	79.78±6.28	54.55±8.39	93.30±5.81	87.59±8.44	48.48±10.75	90.98±3.13	91.96±3.46	91.70±12.81	66.88±11.34	91.60±12.28	95.36±4.80	88.97±2.82	**98.53±4.82**
16	73.92±7.63	59.23±14.55	87.89±9.08	79.48±11.60	53.66±6.84	78.09±12.57	92.94±5.75	96.19±3.56	68.09±15.63	95.15±6.92	95.36±6.26	88.87±6.61	**99.54±3.56**
17	94.00±4.37	66.53±11.04	89.60±8.09	88.93±5.80	86.00±7.88	92.00±10.65	92.27±6.22	87.47±9.21	68.27±11.49	92.80±1.97	97.20±2.31	99.20±0.45	**99.60±1.58**
18	96.00±3.77	96.00±4.62	93.60±5.40	56.40±24.25	79.60±15.49	**100.00±0.00**	99.20±1.69	96.40±3.08	52.80±7.46	97.20±31.39	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
19	91.07±6.13	71.07±14.82	90.00±7.68	94.29±5.63	80.71±12.28	**100.00±0.00**	**100.00±0.00**	96.43±11.54	95.36±5.16	98.57±4.14	98.21±6.80	89.29±10.59	**100.00±0.00**
20	83.52±8.31	88.93±10.94	94.51±6.91	57.95±27.62	85.90±9.49	87.87±7.86	96.07±5.24	95.98±9.00	62.95±16.76	96.14±11.75	99.51±12.20	99.51±13.85	**99.75±0.40**
AA(%)	81.18±1.14	75.14±3.11	88.46±2.41	78.20±4.25	71.78±3.77	81.47±2.25	94.98±1.03	92.73±2.21	71.39±2.98	95.35±3.40	97.13±1.06	95.88±0.70	**98.94±0.61**
OA(%)	85.20±1.24	77.33±2.27	92.74±1.42	89.01±2.02	79.26±2.15	70.65±15.16	93.65±1.78	95.76±0.56	82.09±1.73	97.47±0.89	97.81±0.86	97.51±0.26	**99.28±0.23**
kappa	0.82±0.014	0.73±0.025	0.91±0.017	0.87±0.024	0.75±0.025	0.68±0.158	0.92±0.021	0.95±0.007	0.79±0.021	0.96±0.011	0.97±0.010	0.97±0.003	**0.99±0.003**

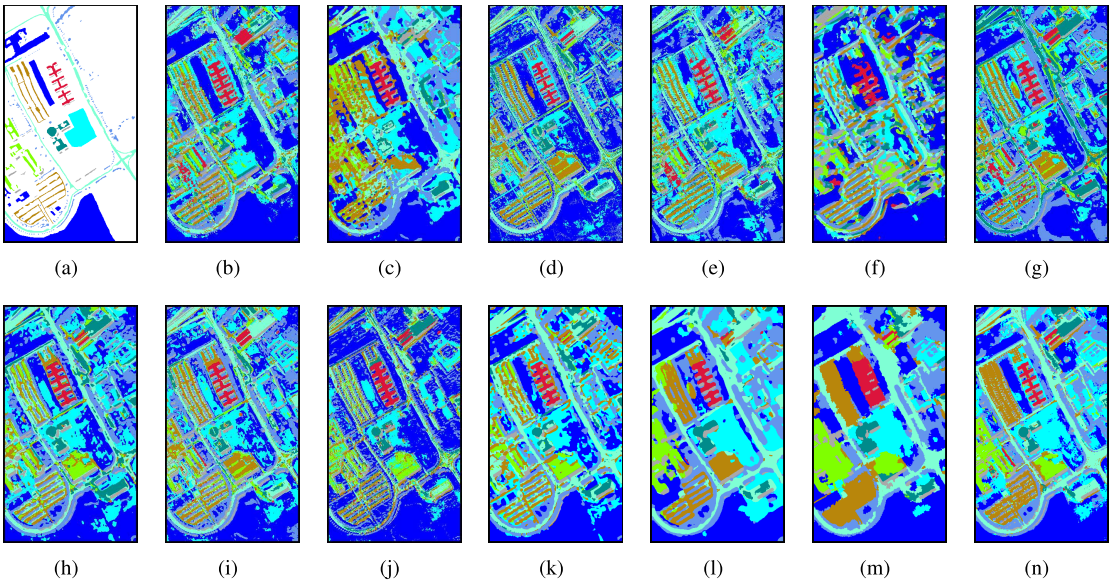


Fig. 11. Classification maps on the PaviaU dataset, including (a) ground truth, (b) Two-CNN, (c) 3DVSCNN, (d) HSGAN, (e) SSLSTMs, (f) SSCL, (g) 3DCAE, (h) MDL4OW, (i) SAGEConv, (j) SpectralFormer, (k) SSFTT, (l) DBCTNet, (m) GiGCN, and (n) SAGT.

SpectralFormer accounts for spectral correlation, it neglects spatially homogeneous features, resulting in weaker performance when there is significant spectral variability. Although SSFTT effectively captures spectral–spatial features using a lightweight Transformer-based structure, its reliance on fixed tokenization and shallow feature extraction limits its adaptability to complex intraclass variations. The DBCTNet adopts a double-branch hybrid design to extract multiscale features with reduced parameters, but the decoupled spatial-spectral processing and simplified attention mechanisms hinder its ability to fully exploit structural dependencies in hyperspectral data. SAGT model surpasses the other comparison methods, highlighting the effectiveness of the proposed backbone.

Classification performance is further contingent on the distribution of image content. This is evidenced by DBCTNet's superior accuracy on the sparsely distributed objects of the PaviaU dataset compared to its performance on the densely clustered YRE scene. Other influencing factors include image resolution, the number of classes, and the degree of interclass similarity. Nevertheless, the proposed method delivers state-of-the-art results across all three benchmark datasets, underscoring its robust and versatile classification capability.

Then, qualitative analysis is also conducted for a comprehensive evaluation. The classification maps with the highest OA among ten random experiments for each method are shown in Figs. 11–13. First, SAGT's classification maps exhibit less noise and closely resemble the ground truth, with clearer boundaries and more complete regions. Furthermore, SAGT demonstrates strong generalization to unlabeled data, with better conformity when compared to false-color HSIs. For example, SAGT clearly and consistently recognizes the shallow sea areas on YRE, while other methods introduce noticeable noise and misclassification in this category. A similar case is observed on the YC dataset, where SAGT correctly

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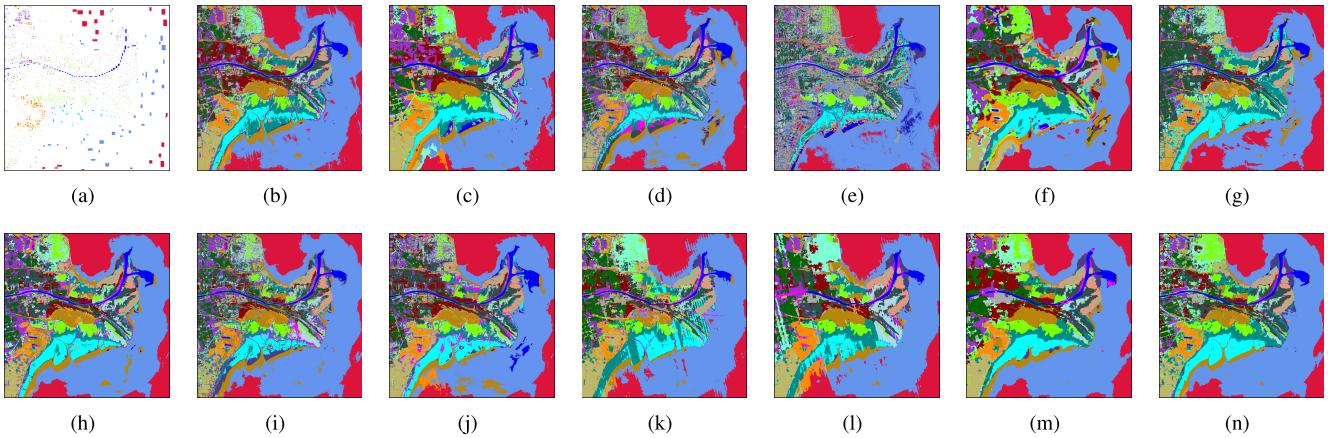


Fig. 12. Classification maps on the YRE dataset, including (a) ground truth, (b) Two-CNN, (c) 3DVSCNN, (d) HSGAN, (e) SSLSTMs, (f) SSCL, (g) 3DCAE, (h) MDL4OW, (i) SAGEConv, (j) SpectralFormer, (k) SSFTT, (l) DBCTNet, (m) GiGCN, and (n) SAGT.

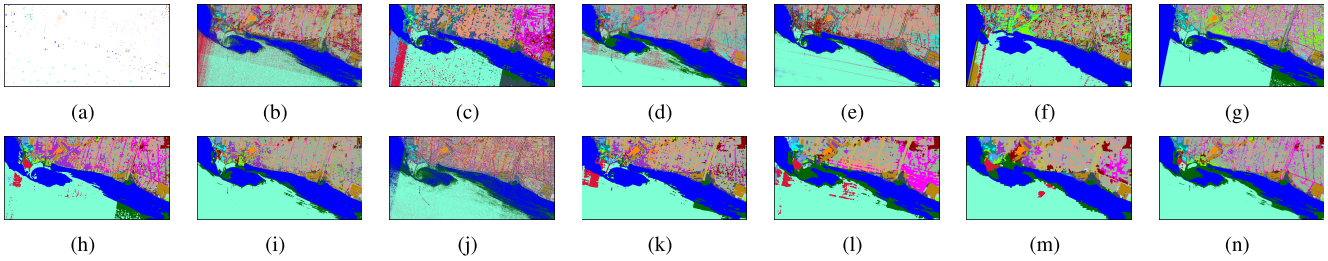


Fig. 13. Classification maps on the YC dataset, including (a) ground truth, (b) Two-CNN, (c) 3DVSCNN, (d) HSGAN, (e) SSLSTMs, (f) SSCL, (g) 3DCAE, (h) MDL4OW, (i) SAGEConv, (j) SpectralFormer, (k) SSFTT, (l) DBCTNet, (m) GiGCN, and (n) SAGT.

differentiates between the sea and offshore water regions. Moreover, our model excels in preserving class boundary clarity and maintaining the integrity of class regions, with an adaptive understanding of both local structures and global relationships. In the false-color image of YC, bare land strips between fields, which appear as narrow and confusing patterns, present distinct spectral characteristics from the surrounding fields. However, many comparison methods tend to misclassify these areas as part of the adjacent fields. In contrast, SAGT successfully distinguishes and classifies them into the correct categories. Therefore, the above results demonstrate that our SAGT model possesses exceptional classification and generalization capabilities across multiscale and diverse scenes.

To evaluate the robustness of models across different sample ratios, we carried out supplementary experiments on PaviaU, YRE, and YC, gradually increasing the number of training samples per class from 5 to 10, as shown in Fig. 14. Our proposed SAGT consistently demonstrates good performance across different training sample sizes. In particular, with only five training samples per class, SAGT achieves performance comparable to the GiGCN, a representative model for few-shot classification. As the number of training samples gradually increases, SAGT consistently improves the classification accuracy and eventually achieves the best performance among all comparison models. This underscores the effectiveness of SAGT in few-shot learning, revealing both its robustness in diverse classification scenarios and its scalability with various training sets.

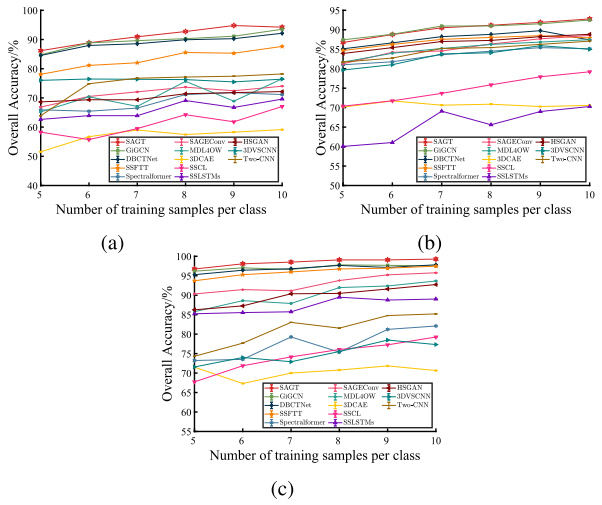


Fig. 14. Classification accuracy versus different numbers of training samples per class on the three HSI datasets. (a) PaviaU. (b) YRE. (c) YC.

E. Efficiency Analysis

Fig. 15 illustrates the parameters of HSI classification methods on the PaviaU, YRE, and YC datasets, which are influenced by the spectral bands, dataset categories, and model architecture. The parameter count of SAGEConv, GiGCN, and SAGT methods is relatively small among all comparison methods. SAGEConv has the fewest parameters, but its performance is not particularly high. In contrast, SAGT has a parameter count comparable to SAGEConv while

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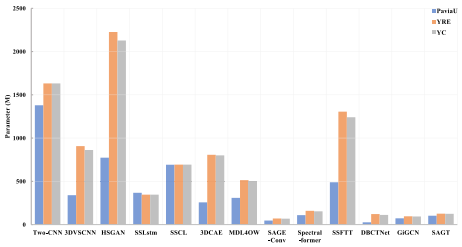


Fig. 15. Number of parameters for HSI classification methods on the PaviaU, YRE, and YC datasets.

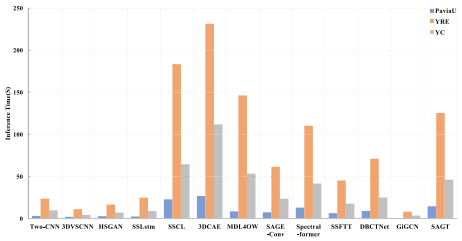


Fig. 16. Total inference time of HSI classification methods on the PaviaU, YRE, and YC datasets.

achieving superior performance, thanks to its SASA and SAP mechanisms. Additionally, SAGT significantly reduces model parameters compared to HSGAN and Two-CNN. Both HSGAN and Two-CNN models employ deep architectures with multiple layers, which inherently increase the parameters. The parameter counts of 3DVSCNN, HSGAN, and 3DCAE significantly increase with the increase of image size, especially in the YRE and YC datasets. Overall, our model learns classification features with effective deformable operators, simplified network architecture, and moderate parameter counts, making it parameter-efficient.

Moreover, we conduct a statistical analysis of total inference times for all classification methods across the three benchmark datasets, as shown in Fig. 16. SAGT achieves feasible performance on the PaviaU, YRE, and YC datasets. The inference cost of SAGT is primarily attributed to the simultaneous learning of node representations and graph structures. Even with its time-consuming dynamic adjustment, the method is still more efficient than models relying on static HSI representations, such as SSCL, 3DCAE, MDL4OW, and SpectralFormer. In summary, SAGT strikes an effective balance among parameter count, inference time, and classification accuracy, thus making it practical for real-world applications.

V. Conclusion

In this article, we propose a novel SAGT to better aggregate homogeneous spatial information, addressing spectral variability and enhancing HSI classification. Specifically, by effectively aggregating homogeneous spectral information, SASA mitigates interference from intraclass spectral variations, thereby ensuring consistent classification of pixels in homogeneous regions. Moreover, by dynamically exploiting spatial topological structures, SAP enhances the prediction accuracy of boundary pixels and improves the fidelity of details of ground objects. Nonetheless, the model's performance is influenced by both the spatial resolution of HSIs and the number of classes. A decrease in resolution, which obscures fine details, coupled with an increase in object categories, leads to a drop in classification accuracy. Overall, the model is particularly well-suited for high-resolution HSIs with complex and variable object structures. Finally, we would like to highlight that the SAGT code is available at https://github.com/ShuGuoJ/SAGT.git., facilitating broader research and application.

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Shuyu Zhang (Member, IEEE) received the B.S. and Ph.D. degrees from the College of Earth Sciences, Zhejiang University, Hangzhou, China, in 2015 and 2020, respectively.

She is currently an Assistant Professor with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China. Her research interests include remote sensing image analysis, ocean data processing, and deep learning.

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Shuguo Jiang received the B.E. degree from Xiamen University of Technology, Xiamen, China, in 2020, and the M.E. degree from Shenzhen University, Shenzhen, China, in 2023. He is currently pursuing the Ph.D. degree with the School of Computer Science, Wuhan University, Wuhan, China.

His research interests include remote sensing and multimodality interpretation.



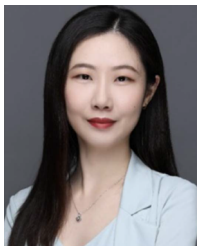
Wenlong Yin received the B.E. degree in medical information engineering from Jiangxi University of Chinese Medicine, Nanchang, China, in 2023. He is currently pursuing the master's degree in computer technology with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China.

His research interests include hyperspectral image classification and deep learning.



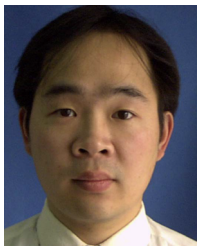
Weixi Wang received the Ph.D. degree in geodesy and survey engineering from Liaoning Technical University, Huludao, China, in 2007.

He was a Post-Doctoral Fellow with Wuhan University, Wuhan, China, in 2013. He is currently a Full Professor with Shenzhen University, Shenzhen, China. He is mainly engaged in research on building information modeling (BIM), 3-D geographical information systems (GISs), and virtual geographic environments (VGEs).



Meng Xu (Member, IEEE) received the B.S. and M.E. degrees in electrical engineering from the Ocean University of China, Qingdao, China, in 2011 and 2013, respectively, and the Ph.D. degree from the University of New South Wales, Canberra, ACT, Australia, in 2017.

She is currently an Associate Professor with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China. Her research interests include cloud removal and remote sensing image processing.



Jiasong Zhu received the B.E. degree from the Huazhong University of Science and Technology, Wuhan, China, in 1997, the M.E. degree from Wuhan University, Wuhan, in 2003, and the Ph.D. degree from The University of Hong Kong, Hong Kong, in 2008.

He is currently a Full Professor with the College of Civil Engineering, Shenzhen University, Shenzhen, China. His research interests include multisensor integration and data fusion, high-resolution image processing, and geographic information systems (GIS) applications in urban planning and transportation.



Sen Jia (Senior Member, IEEE) received the B.E. and Ph.D. degrees from the College of Computer Science, Zhejiang University, Hangzhou, China, in 2002 and 2007, respectively.

Since 2008, he has been with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China, where he is currently a Full Professor. His research interests include remote sensing image processing, signal and image processing, and machine learning.

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